

AI Transformation Portal

Architecture & Data Flow — Technical Documentation

المنصة الحكومية لتخطيط ومتابعة مشروع الذكاء الاصطناعي المساعد

Document 1 of 2 · Prepared for the IT team (staging deployment, DB linking, go-live)

Version 1.0 · July 2026 · Companion documents: HANDOVER.md, DEPLOYMENT.md, Roles & Access (PDF)

1. System overview

The portal is a **Next.js 14** application (App Router, TypeScript, RTL Arabic UI) backed by **PostgreSQL 16** through **Prisma ORM**. Users sign in with **UAE PASS** (OIDC). An optional internal, OpenAI-compatible model powers the AI review step. One codebase serves all four roles; every screen derives from a single role-scoped view-model, so what a user can see is decided in one place.

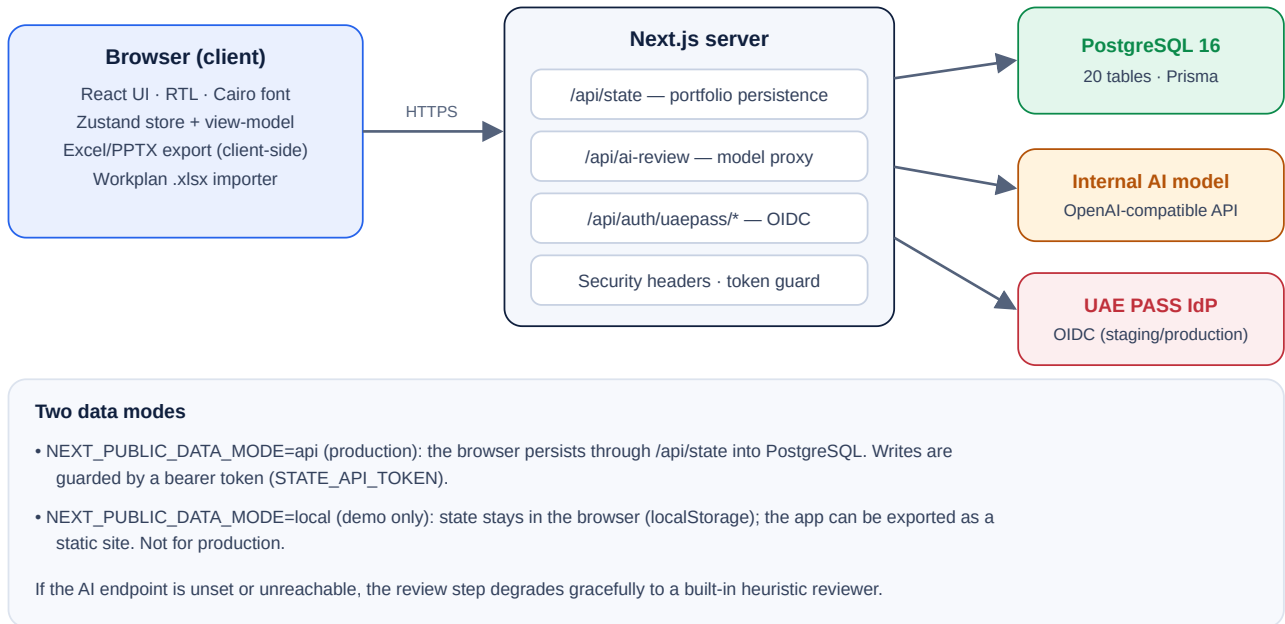


Fig. 1 — Runtime architecture

2. Deployment architecture (Kubernetes)

Manifests are provided in **k8s/**. The application image is built from the repository **Dockerfile**; database schema changes run as a one-shot migration **Job** before each rollout.

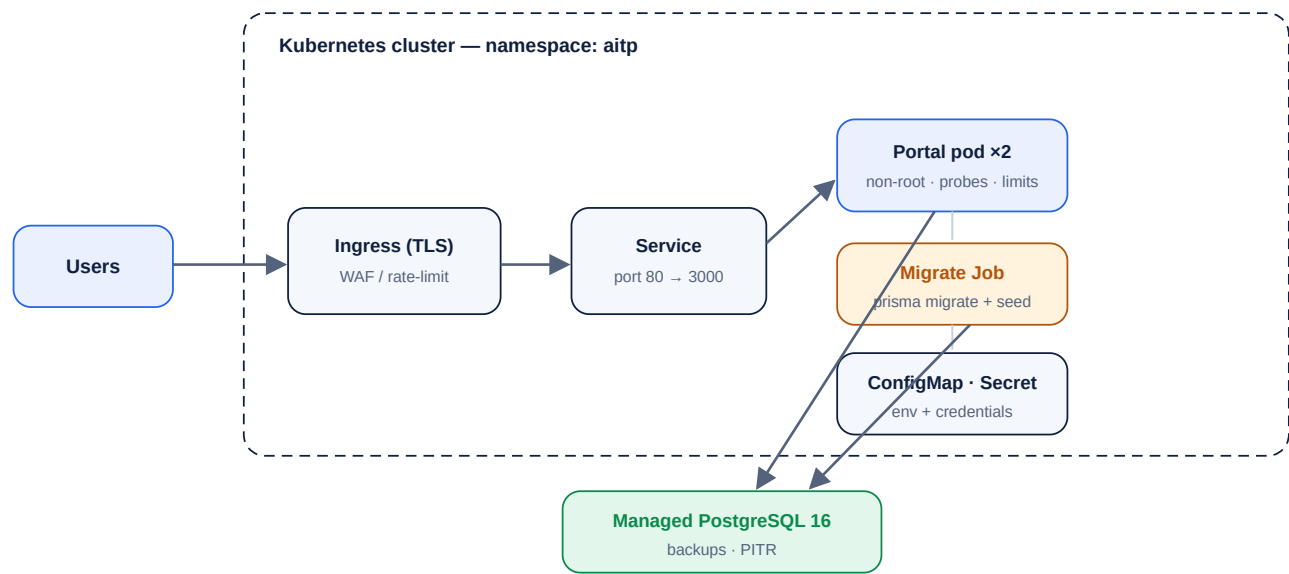


Fig. 2 — Kubernetes deployment (k8s/ manifests)

Release flow

#	Step	Command / artifact
1	Build & push image	<code>docker build -t REGISTRY/aitp-portal:TAG .</code>
2	Apply namespace, config, secret	<code>k8s/namespace.yaml · configmap.yaml · secret (from secret.example.yaml)</code>
3	Migrate + seed reference data	<code>k8s/migrate-job.yaml (idempotent)</code>
4	Roll application	<code>k8s/deployment.yaml · service.yaml · ingress.yaml</code>

3. Data model

Schema source of truth: **prisma/schema.prisma** + migrations **0001–0005** (verified on a fresh PostgreSQL 16). The seed loads **reference data only** — users, items, launch plans and all transaction tables start empty.

Group	Tables	Notes
Reference	streams, entities, program_phases, exec_batches, settings	5 streams in official order with the predefined heads (رؤساء المسارات); 35 federal entities; المراحل; approved committee budget (100M AED).
Identity	users, entity_reps, stream_owners	users is empty by design — IT creates accounts and maps UAE PASS sign-ins (role: entity coord path ai, plus stream/entity links).
Portfolio	items, item_launch_plans, launch_plans, launches, item_launches, exec_checklist_items, sub_milestones	Items carry the execution budget ; a launch plan's execution budget is derived (sum of attached items) and its launch_budget is informational only.
Workflow / audit	log_entries, nominations, fundings, funding_cancellations	Every action is logged with actor + timestamp; nominations and funding decisions are first-class rows.
Misc	app_state	Serialized UI-state snapshot used by /api/state.

Cost model (important)

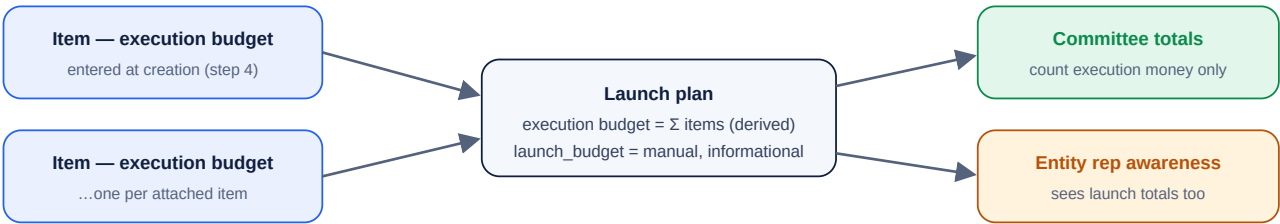


Fig. 3 — Cost derivation: items feed plans; funding counts execution money only

4. Data flow — sign-in (UAE PASS OIDC)

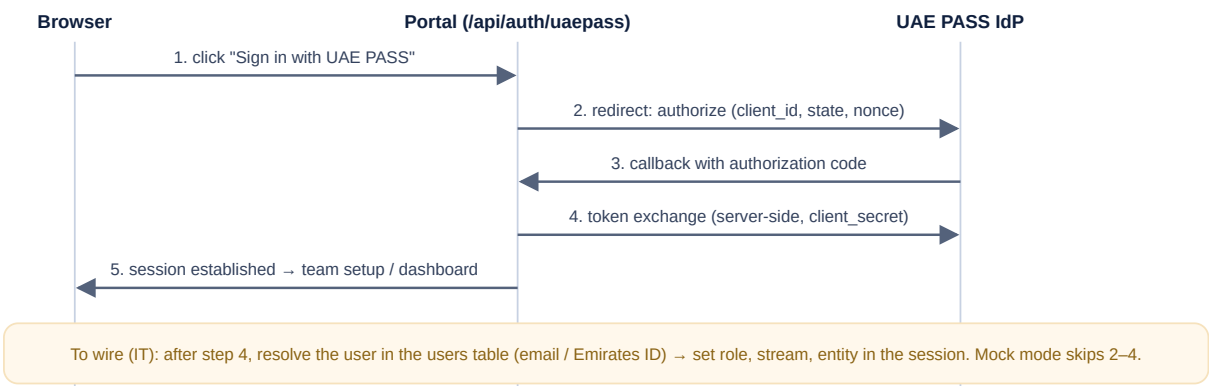


Fig. 4 — Authentication sequence

5. Data flow — item lifecycle

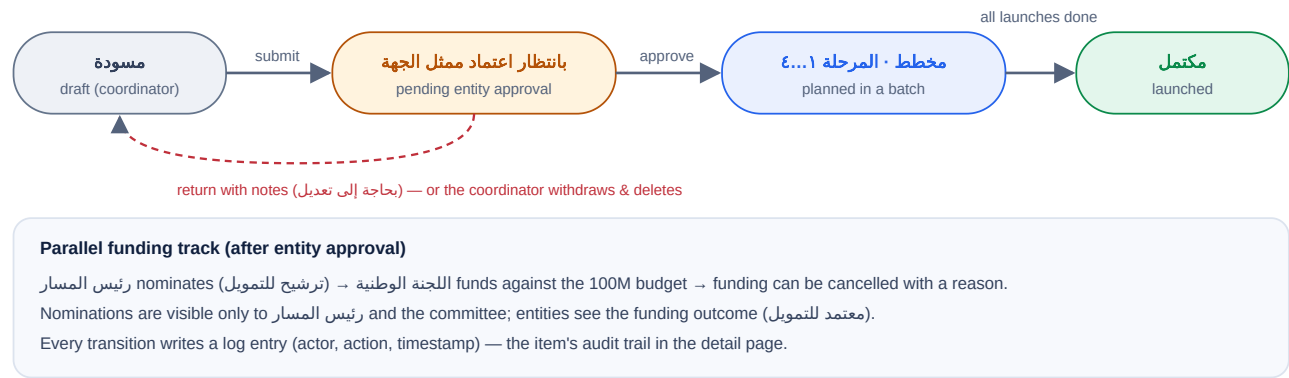


Fig. 5 — Lifecycle state machine + funding track

6. Data flow — bulk import (workplan file)

The coordinator uploads the **official خطة العمل Excel export** that entities already produce (the same file from the existing workplan system). The importer reads every sheet and maps the legacy vocabulary automatically.

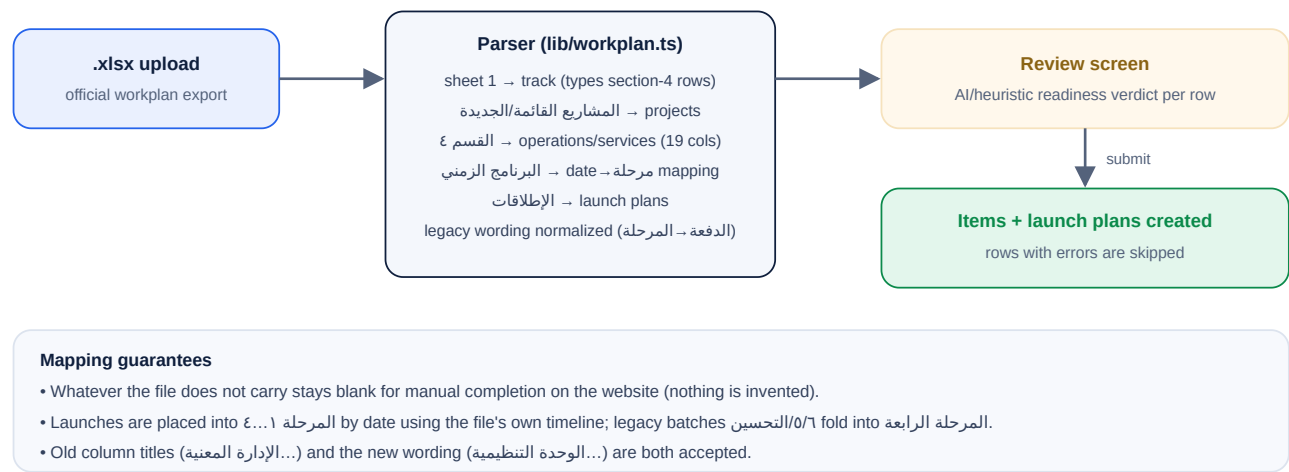


Fig. 6 — Bulk import pipeline

7. Data flow — persistence & exports

Flow	Path	Notes
State write	UI action → store → POST /api/state → PostgreSQL	Bearer-token guarded; every workflow action also appends a log entry.
State read	App load → GET /api/state → store hydrate	Role scoping is applied in the view-model on every render.
AI review	UI → POST /api/ai-review → internal model	Server-side proxy; heuristic fallback if the model is unavailable.
Excel / PowerPoint export	Client-side generation (exceljs / pptxgenjs)	Exports the current role-scoped list only — a user can never export data they cannot see.